Protein Fouling of Virus Filtration Membranes: Effects of Membrane Orientation and Operating Conditions

Zeeshan H. Syedain, David M. Bohonak, and Andrew L. Zydney*

Department of Chemical Engineering, The Pennsylvania State University, University Park, Pennsylvania 16802

The capacity of virus filters used in the purification of therapeutic proteins is determined by the rate and extent of membrane fouling. Current virus filtration membranes have a complex multilayer structure that can be used with either the skin-side up or with the skin-side facing away from the feed, but there is currently no quantitative understanding of the effects of membrane orientation or operating conditions on the filtration performance. Experiments were performed using Millipore's Viresolve 180 membrane under both constant pressure and constant flux operation with sulfhydryl-modified BSA used as a model protein. The capacity with the skin-side up was greater during operation with constant flux and at low transmembrane pressures, with the flux decline or pressure rise due primarily to osmotic pressure effects. In contrast, data obtained with the skin-side down showed a slower, steady increase in total resistance with the cumulative filtrate volume, with minimal contribution from osmotic pressure. The capacity with the skin-side down was significantly greater than that with the skin-side up, reflecting the different fouling mechanisms in the different membrane orientations. These results provide important insights for the design and operation of virus filtration membranes.

Introduction

Virus filtration has become an important component of the overall viral clearance strategy in the production of biopharmaceutical products derived from human or animal origin. Virus filtration provides a robust, size-based removal mechanism that compliments other physical removal techniques (e.g., chromatography) and a variety of inactivation methods (irradiation, temperature, pH, and solvent-detergent mixtures). Burnouf and Radosevich (*1*) have reviewed the use of virus filtration in the preparation of plasma-derived products, including coagulation factors, immunoglobulins, transferrins, and protease inhibitors. Overviews of the use of membranes for virus filtration in the biopharmaceutical industry have been provided by Carter and Lutz (*2*) and Levy et al. (*3*).

Currently Millipore Corporation, Pall Corporation, and Asahi Kasei manufacture commercial membranes specifically designed for virus removal applications. These membranes typically have a complex multilayer structure to ensure that there are no defect paths, which is absolutely essential to obtain the high levels of virus removal (greater than 99.9%) needed for bioprocessing applications (*1*, *4*). Virus filters are used in both normal flow filtration (NFF), which is also referred to in the literature as dead-end or direct flow filtration, and in cross-flow or tangential flow filtration (TFF). Although tangential flow can provide higher filtrate flux by reducing concentration polarization, the current trend is toward normal flow filtration due to the simpler design and reduced costs.

One of the critical issues in the design of normal flow virus filtration systems is membrane fouling. Fouling can significantly limit the filter capacity, which is defined as the total volume of filtrate that can be collected before the filtrate flux declines to an unacceptable value during constant pressure operation (or before the transmembrane pressure increases to its maximum value during constant flux operation). Specific limits on the maximum pressure are determined by the membrane manufacturer on the basis of mechanical constraints on the module or membrane or by the end-user on the basis of other process limitations.

Since virus filtration is usually performed near the end of the downstream purification, the process stream contains primarily the protein of interest with low levels of contaminants and very low levels of virus. Thus, the dominant foulant of most virus filters will be the protein product. There have been a number of comprehensive reviews of the mechanisms governing protein fouling during microfiltration and ultrafiltration (*5*, *6*), but these results are difficult to apply to virus filtration. The complex multilayer structure of the virus filtration membranes feature selective pores that are only slightly larger than the size of the proteins, allowing proteins to pass through the membrane and into the filtrate (protein loss is typically much less than 10%). Virus filters thus lie between conventional ultrafiltration membranes, which are fully retentive to most proteins, and conventional microfiltration membranes, which have pores that are an order of magnitude or more greater than the size of the proteins of interest.

Despite the importance of protein fouling during virus filtration, there have been relatively few studies of this phenomenon. Brough et al. (*4*) evaluated the capacity of Viresolve NFR membranes, which have a triple layer composite structure, in normal flow filtration. The capacity for a 5 g/L BSA solution was more than 8000 L/m^2, whereas that for a 180 kD protein was approximately 4000 L/m^2, with the capacities estimated using the $V_{\max}$ method (*7*). However, no data were provided on the specific conditions or the mechanism of fouling. Higuchi et al. (*8*) obtained data for the filtrate flux during constant pressure filtration of 5 g/L γ-globulin solutions through Planova 35N membranes. The capacity in the absence

* To whom correspondence should be addressed. Ph: 814-863-7113. Fax: 814-865-7846. Email: zydney@engr.psu.edu.

10.1021/bp050350v CCC: $33.50
Published on Web 05/11/2006

of salt was only 10 L/m^2 as a result of the rapid fouling under these conditions. The capacity was significantly larger in the presence of salt (*8*) or upon treatment with DNase (*9*), which the authors attributed to the dissociation of large complexes that formed between the γ-globulin and trace amounts of DNA present in the feed solution. These complexes were thought to block the pores in the multilayer structure of the Planova membranes. Subsequent work by Higuchi et al. (*10*) showed the importance of large protein aggregates on the flux decline with both albumin and γ-globulin solutions.

The objective of this study was to quantify the effects of operating mode on the capacity and fouling characteristics of the Viresolve 180 membranes. These membranes are made of hydrophilized polyvinylidene fluoride and have a composite asymmetric structure with a 1 μm thick skin layer on top of a fairly dense intermediate layer that is supported on a more open microfiltration membrane (*11*). The pores in the Viresolve 180 membrane are large enough to allow nearly complete passage of proteins 180 kD in size but effectively retain viruses as small as 25 nm. Experiments were performed using both constant pressure and constant flux operation, with bovine serum albumin (BSA) used as a model protein foulant due to the extensive previous work on BSA fouling during both ultrafiltration (*12, 13*) and microfiltration (*6*, *14*−*16*). Data were also obtained with the membrane oriented so that the skin was either toward or away from the feed to examine the effect of membrane orientation on the fouling characteristics.

Materials and Methods

All experiments were conducted in 10 mL Amicon stirred cells (model 8010, cat. no. 5121, Millipore Corp.) using Viresolve 180 membranes provided by Millipore Corporation (Lot V180.H013001 G3-101). Data were obtained using solutions of cysteinylated bovine serum albumin (cys-BSA from Serological Proteins Inc., Lot 1038). The cys-BSA has been chemically treated by the manufacturer to block the free sulfhydryl groups present in BSA. Previous studies by Kelly and Zydney (*17*) have demonstrated that this chemical modification reduces the extent of BSA dimerization and aggregation, providing a more reproducible protein solution with reduced fouling characteristics. In addition, the behavior of the cys-BSA is likely to be more representative of purified recombinant proteins, which typically have much lower degrees of dimerization than conventional BSA solutions.

Protein powders were dissolved in phosphate buffered saline solution (PBS) prepared by dissolving 0.03 M KH_2PO_4 (EM Science or J. T. Bakers), 0.03 M $Na_2HPO_4 \cdot 7H_2O$ (J. T.Bakers), and 0.015 M NaOH (EM Science) in deionized water obtained from a Nanopure Diamond water purification system (Barnstead/Thermodyne) with a minimum resistivity of 18 MΩ-cm. The pH was measured using a 420A pH meter (Thermo Orion) and adjusted to 7.4 by addition of NaOH or phosphoric acid as needed. PBS solutions were prefiltered through an 0.2 μm pore size Supor-200 filter (Pall Corp.) followed by a 100 kD Biomax ultrafiltration membrane (Millipore Corp.) to remove any undissolved salts or particulates. Buffers used in the preparation of samples for scanning electron microscopy additionally contained 0.5 g/L sodium azide as a bacteriostat.

Protein Filtration. The virus filtration membranes were cut into 25-mm diameter circular disks from large membrane sheets using a specially designed cutting device. The membrane disks were soaked in a PBS solution for approximately 30 min. The wet membrane was then placed in the stirred cell with either the skin-side up or with the membrane substructure positioned to face the feed solution. The membrane was supported on a Tyvek spacer that was placed immediately beneath the membrane at the base of the stirred cell. The membrane was then flushed with approximately 70 L/m^2 of a 100 kD-filtered PBS solution (for approximately 30 min). The stirred cell was then carefully emptied and refilled with a BSA solution. The device was then connected to a pressurized reservoir containing additional protein solution.

Experiments at constant filtrate flow rate were performed using a DynaMax peristaltic pump (model RP-1, Rainin Instrument Co., Inc.) that was connected to the outlet permeate tube from the base of the stirred cell. The solution reservoir was pressurized to 172 kPa (25 psi) to maintain a positive gauge pressure, with the transmembrane pressure evaluated using a PX26 Series pressure transducer (Omega Engineering, Inc.). Small filtrate samples were collected at regular time intervals, with the flow rate measured by timed collection using a digital balance. The protein concentration in the collected filtrate was evaluated spectrophotometrically on the basis of the natural sample absorbance at 280 nm using a UV−vis spectrophotometer (Shimadzu).

A similar approach was used for the constant pressure experiments, except the pump on the outlet line was removed so that the flow was simply driven by the pressurization of the feed solution reservoir. Filtrate flow rates were evaluated by timed collection throughout the experiment. All experiments were performed at room temperature (22 ± 2 °C), with all solutions freshly prepared (less than 48 h for the PBS and less than 2 h for the BSA).

Scanning electron micrographs of membrane surfaces were obtained at 3.5 kV using a Hitachi S-3000H scanning electron microscope available in the Materials Characterization Laboratory at the Pennsylvania State University. Images were obtained with membranes that were simply soaked in 1 g/L cys-BSA solutions at 4 °C for 12 h and with membranes that were fouled during protein filtration. The latter membranes were gently rinsed with buffer and then washed by filtration using at least 50 L/m^2 of PBS. The fouled membranes were soaked in buffer for at least 12 h at 4 °C, followed by a solution containing 2.5% glutaraldehyde for 30 min to fix the protein deposit. The membranes were then washed successively in solutions containing 50%, 70%, 85%, 95%, and 100% ethanol (twice) to remove water. The samples were further dehydrated and dried with CO_2 using a Bal-tec CPD-030 critical point dryer (Techno Trade). The dried membranes were subsequently sputtered with a 5-nm layer of gold and imaged at a working distance of approximately 11 mm.

Results and Analysis

Skin-Side Up. Typical experimental data for the filtrate flux, J_v, during the constant pressure filtration of a 1 g/L solution of cysteinylated-BSA through the Viresolve 180 membrane with the skin-side up (facing the feed) are shown in Figure 1. Data are shown for three separate experiments performed at transmembrane pressures of 69, 103, and 172 kPa (corresponding to 10, 15, and 25 psi), with the results plotted as a function of the cumulative filtrate volume. The filtrate concentration in these experiments was essentially equal to 1 g/L throughout these unstirred filtration runs, indicating that there was no significant protein retention by the Viresolve 180 membrane when used in normal flow filtration (consistent with the relatively large pore size). The initial filtrate flux is greatest at the highest pressure of 172 kPa, but the situation reverses at longer times, with the highest filtrate flux obtained for the run with the lowest applied

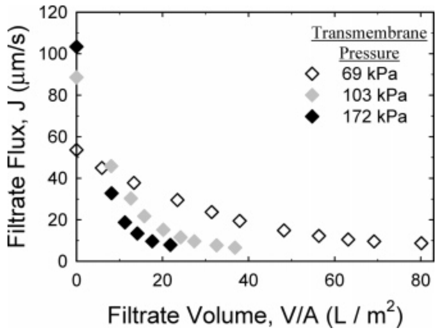

Figure 1. Filtrate flux as a function of the cumulative filtrate volume per unit membrane area for the Viresolve 180 membrane in the skin-side up orientation for operation at constant transmembrane pressure.

pressure. The capacity, defined as the cumulative filtrate volume when the flux drops to 20% of the clean membrane value, was only 9 L/m^2 at 172 kPa, which is significantly less than the 17 L/m^2 obtained at a pressure of 103 kPa and only 15% of the 57 L/m^2 capacity at 69 kPa. Similar differences were seen using other definitions of the capacity, e.g., the cumulative filtrate volume obtained when the flux declines to 10% or 30% of the initial value.

Additional insights into the fouling phenomena were obtained using scanning electron microscopy. Figure 2 shows SEM images of the upper surface of the membrane skin, both for a membrane that was soaked overnight in a 1 g/L protein solution and for a membrane used to filter a 1 g/L cys-BSA solution at a constant pressure of 103 kPa until a 90% flux decline was observed. The membrane soaked in BSA shows a small number of submicron particles, which are likely undissolved salts or some protein aggregates. The fouled membrane shows a much greater amount of surface coverage, with both micron and submicron material on the membrane. These particles were deposited on the membrane as a result of the convective flow; they are not seen when protein transport occurs by a purely diffusive mechanism as in the simple adsorption experiment. However, most of the membrane pores remain available for filtration, suggesting that irreversible fouling is not the dominant factor governing the flux decline during constant pressure operation.

The relatively small effect of the fouling deposit was confirmed by measurements of the membrane resistance:

$$R_{\mathrm{m}} = \frac{\Delta P}{\mu J_{\mathrm{v}}} \tag{1}$$

Data were obtained for both the clean membrane and then for the same membrane after protein filtration. In both cases, the resistance was determined using protein-free phosphate buffer to evaluate the flux, where μ is the viscosity of the buffer solution. The resistance of the fouled membrane was only double that for the clean membrane after an extended filtration, i.e., after the flux had dropped to approximately 6% of its initial clean-membrane value, providing conclusive evidence that the flux decline seen in Figure 1 was not due primarily to irreversible fouling.

Experimental data for the increase in transmembrane pressure, ΔP, as a function of the cumulative filtrate volume for experiments performed at constant filtrate flux are presented in Figure 3. Data are shown for three separate experiments conducted at constant fluxes of 2.5×10^{-5}, 2.9×10^{-5}, and 3.2×10^{-5} m/s. The initial transmembrane pressure increases with increasing flux as expected, with values corresponding to an essentially constant (pressure-independent) membrane resistance. The pressure increases slowly during the initial phase of the filtration but then begins to increase quite rapidly after about $V = 70$ L/m^2 (corresponding to $t = 11$ min) at $J_v = 3.2 \times 10^{-5}$ m/s. This rapid increase in pressure occurs after a considerably longer filtration time for the runs at lower flux, with the pressure rise at $J_v = 2.5 \times 10^{-5}$ m/s occurring around $V = 300$ L/m^2 ($t = 200$ min). The net result is a dramatic increase in capacity for the runs at lower filtrate flux. For example, if the maximum system pressure is taken as 138 kPa (20 psi), the capacity for the experiment at $J_v = 3.2 \times 10^{-5}$ m/s is only 82 L/m^2 compared to 173 L/m^2 at a flux of 2.9×10^{-5} m/s and 317 L/m^2 at a flux of 2.5×10^{-5} m/s. The capacities during filtration using both constant pressure and constant filtrate flux are summarized in Table 1 and discussed in more detail subsequently.

To obtain additional insights into the fouling mechanisms that govern the behavior during constant flux filtration, an experiment was performed in which stirring was initiated after the system pressure had attained a value of 172 kPa. Results for the transmembrane pressure and normalized filtrate concentration are shown in the top and bottom panels of Figure 4. The flux was maintained at a constant value of 3.2×10^{-5} m/s throughout the experiment. The transmembrane pressure begins to drop immediately after the stirring was started, with the pressure decreasing very rapidly by about a factor of 2. The stirring also caused a reduction in the normalized filtrate concentration, with the sieving coefficient decreasing to a value around 0.5 shortly after initiation of the stirring. These results strongly suggest that there is a significant amount of concentration polarization in this system, with the stirring reducing the protein concentration in the vicinity of the membrane by increasing the rate of back mass transport. This reduction in the protein concentration near the membrane causes a drop in the filtrate concentration since the membrane allows a constant fraction of the protein at the membrane surface to pass into the filtrate. In addition, this decrease in local protein concentration leads to a reduction in the osmotic pressure difference across the membrane, lowering the transmembrane pressure drop needed to maintain a constant flux as seen in Figure 4. The subsequent increase in the protein concentration in the filtrate solution at 92 L/m^2 is due to the accumulation of rejected protein in the stirred cell and does not reflect any significant changes in the degree of concentration polarization or in the actual sieving properties of the membrane.

A simple stagnant film analysis can be used to estimate the degree of concentration polarization in the stirred system. The protein concentration at the membrane surface, C_{w}, can be calculated as

$$C_{\mathrm{w}} = C_{\mathrm{f}} + (C_{\mathrm{b}} - C_{\mathrm{f}})\cdot\exp\left(\frac{J_{\mathrm{v}}}{k}\right) \tag{2}$$

where C_{f} and C_{b} are the protein concentrations in the filtrate and in the bulk solution, respectively. The solute mass transfer coefficient, k, in the presence of stirring was calculated as 5.4×10^{-6} m/s using the correlation presented by Smith et al. (*18*), with this value consistent with an independent estimate by Opong and Zydney (*19*). The sieving data obtained after initiation of the stirring give a value of $C_{\mathrm{w}} = 150$ g/L, which corresponds to an actual sieving coefficient of 0.004. This implies that the protein concentration in the vicinity of the membrane in the unstirred system, where the filtrate concentration was approximately 1.2 g/L, is on the order of 300 g/L. The change in the osmotic pressure upon initiation of the stirring is thus around 95 kPa using the experimental correlation for the osmotic pressure of bovine serum albumin solutions obtained

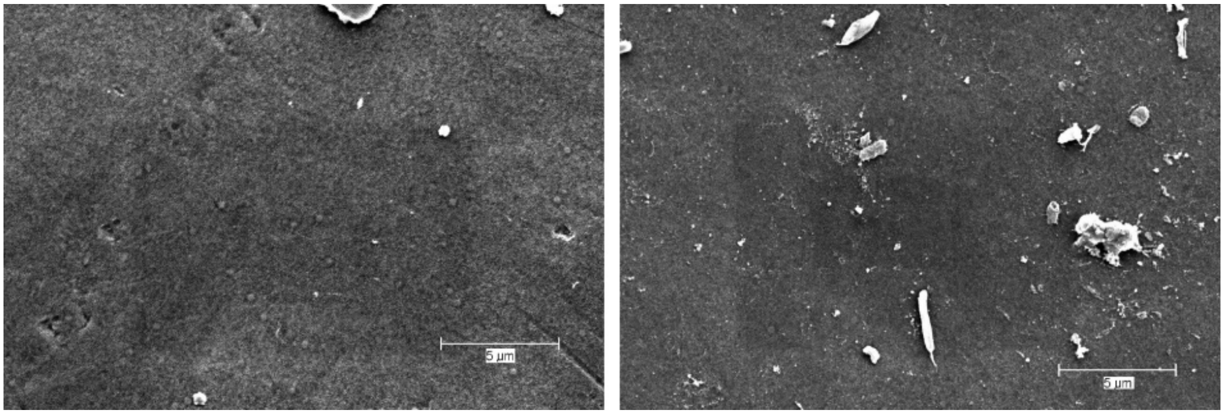

Figure 2. Scanning electron micrographs of the upper (skin side) surface of the Viresolve 180 membrane after soaking in a 1 g/L cys-BSA solution (left panel) and after fouling during constant pressure filtration at 103 kPa (right panel).

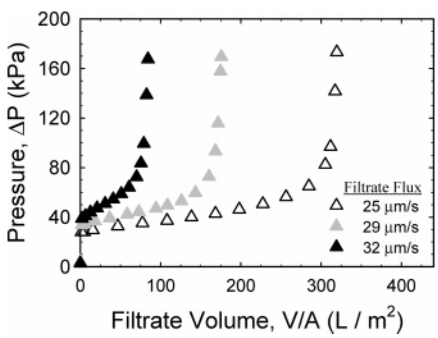

Figure 3. Transmembrane pressure as a function of the cumulative filtrate volume per unit membrane area for the Viresolve 180 membrane in the skin-side up orientation for operation at constant filtrate flux.

Table 1. System Capacity for Different Operating Modes and Membrane Orientations

	system capacity (L/m^2)	
	skin-side up	skin-side down
	Constant Pressure	
69 kPa	41	224
103 kPa	13	247
172 kPa	7	230
	Constant Flux	
25 μm/s	317	
29 μm/s	173	>430
32 μm/s	82	
48 μm/s		>132
62 μm/s		94
92 μm/s		29

by Vilker et al. (*20*). This calculated change in osmotic pressure is in good agreement with the experimental results for the pressure drop upon initiation of stirring (top panel, Figure 4), providing strong evidence that osmotic pressure effects provide the dominant contribution to the pressure profiles obtained during constant flux filtration with the skin-side up. This is also consistent with the relatively small increase in resistance of the fouled membrane ($R_{\mathrm{fouled}}/R_{\mathrm{clean}} \approx 1.6$ for the constant flux experiments).

Skin-Side Down. Although virus filtration membranes were originally developed for use in the skin-side up orientation, typical of conventional ultrafiltration membranes, recent applications of normal flow virus filtration have employed the skin-side down orientation. Jonsson and Wenten (*22*) originally described the use of membranes in "reverse orientation," i.e., with the skin oriented away from the feed, for tangential flow

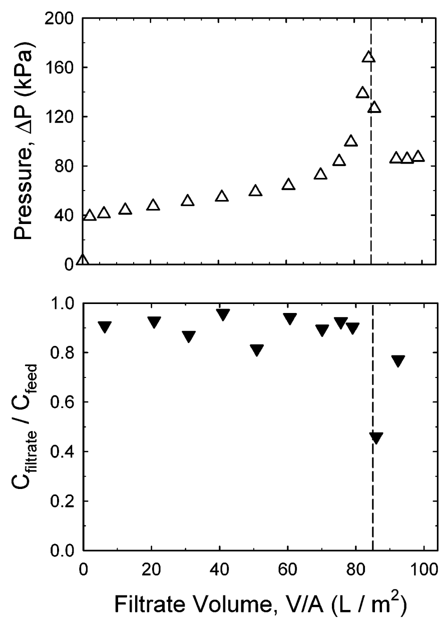

Figure 4. Transmembrane pressure (top panel) and normalized filtrate concentration (bottom panel) for the Viresolve 180 membrane in the skin-side up orientation for operation at a constant filtrate flux of 32 μm/s. Stirring at 600 rpm was initiated at a filtrate volume of 85 L/m^2.

microfiltration, although the focus of that work was on concentration polarization effects on solute retention.

Figure 5 shows experimental data for the filtrate flux as a function of the cumulative filtrate volume for the constant pressure filtration of 1 g/L cys-BSA solutions through the Viresolve 180 membrane with the skin side oriented away from the feed solution. The feed was not stirred in these experiments. The initial flux in the skin-side down orientation is somewhat higher than that obtained with the skin-side up, a phenomenon discussed previously by Bohonak and Zydney (*21*). The flux declines with filtrate volume during filtration at all three pressures, with the magnitude of the flux decline being slightly greater at the higher pressures. In contrast to the data in Figure 1 for the membrane in the skin-side up orientation, the filtrate flux at the higher pressures remains higher than that at the lower pressures throughout the experiment. The net result is that the filter capacity is a weak function of the transmembrane pressure with a value of 230 $\pm$ 20 L/m^2 for pressures ranging from 69 to 172 kPa. For these skin-side down experiments, the capacity at constant pressure was defined as the cumulative filtrate volume at which the flux decreases to 30% of its initial value; it was not possible to use the volume at 10% of the initial flux

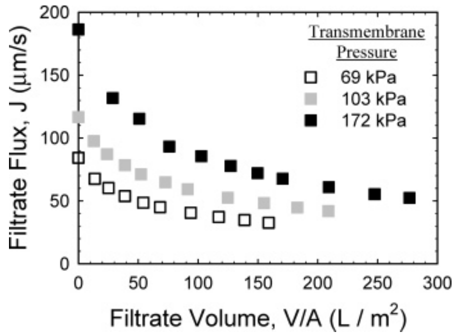

Figure 5. Filtrate flux as a function of the cumulative filtrate volume per unit membrane area for the Viresolve 180 membrane in the skin-side down orientation for operation at constant transmembrane pressure.

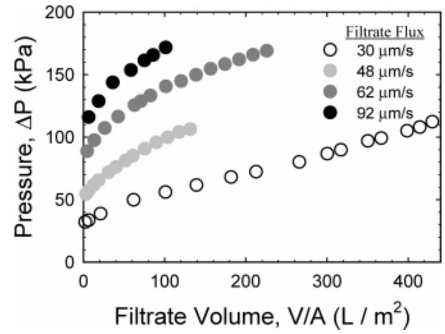

Figure 6. Transmembrane pressure as a function of the cumulative filtrate volume per unit membrane area for the Viresolve 180 membrane in the skin-side down orientation for operation at constant filtrate flux.

due to the fairly low rate of fouling seen in these experiments. If the capacity were instead defined on the basis of a minimum filtrate flux of 50 μm/s, then the capacity would actually increase with increasing transmembrane pressure, going from 50 L/m^2 at 69 kPa to 300 L/m^2 at 172 kPa.

Corresponding data for the transmembrane pressure profiles during constant flux filtration using the skin-side down orientation are shown in Figure 6. Experiments were performed at flux values of 3.0, 4.8, 6.2, and 9.2 $\times$ 10^{-5} m/s, which span a considerably larger range than the flux values used with the skin-side up (Figure 3). The pressure increases with time as expected, with the fastest increase at the highest filtrate flux. The increase in pressure is fairly uniform throughout these filtration experiments, with the data being slightly concave down particularly at the higher flux. This behavior contrasts with the results for the skin-side up orientation (Figure 3) where the data were strongly concave up, with the pressure rising rapidly after some critical filtration volume. Regardless of the filtration endpoint, i.e., the choice of maximum pressure, the filter capacity in the skin-side down orientation increases with decreasing flux, although the dependence on the flux is much weaker than that observed for the skin-side up orientation (Table 1).

Measurements of the hydraulic resistance of the fouled membranes, again obtained using protein-free phosphate buffer solution, showed a significant increase in resistance. For example, the resistance increased from 1.2 to 3.8 $\times$ 10^{12} m^{-1} after a constant pressure filtration at ΔP = 103 kPa conducted until J_v/J_o = 0.14. The 3-fold increase in membrane resistance in the skin-side down orientation is more consistent with the observed decrease in filtrate flux than the results obtained with the skin-side up. Scanning electron micrographs obtained with the fouled membrane showed little fouling on the external surface of the membrane when used in the skin-side down orientation, suggesting that the increase in resistance arises from foulant deposition within the large pores of the membrane substructure when the skin side is oriented down (away from) the feed. The increase in membrane resistance can be divided by the total protein mass per area fed to the membrane throughout the entire filtration. In the skin-side down orientation, this ratio has a value of 6.8 $\times$ 10^9 m/kg, roughly one-fifth of the value of 3.6 $\times$ 10^{10} m/kg obtained when the skin was upstream. These data indicate that irreversible fouling by a given mass of protein in the skin-down orientation has a considerably smaller effect on the membrane permeability, since the protein deposits throughout the membrane thickness rather than in a tight layer on the skin side where the membrane pores are smallest.

The capacity data for experiments performed with the different membrane orientations and for constant flux and constant pressure operation are summarized in Table 1. The capacity for the runs with constant flux was defined as the total filtrate volume collected per unit membrane area up until a pressure of 138 kPa (20 psi), while the capacity for the runs with constant pressure was defined as the filtrate volume obtained until the filtrate flux had decayed to 30% of its initial value for both the skin-side up and skin-side down orientations. The capacity was essentially independent of applied pressure with the skin-side down in sharp contrast to the nearly 6-fold reduction in capacity with increasing pressure seen in the skin-side up orientation. The capacity during constant flux operation decreases with increasing flux in both the skin-side down and skin-side up orientations, although the dependence on the flux is much less pronounced with the skin-side down.

The data in Figures 5 and 6 are replotted in the top panel of Figure 7 as the total resistance:

$$R_{\text{total}} = \frac{\Delta P}{\mu J_{\text{v}}} \tag{3}$$

The data appear to collapse to a single curve when plotted in this manner, with only small differences between results at different pressures/fluxes or in the different operating modes (constant pressure versus constant flux). This behavior implies that the fouling in this system is determined completely by the mass of protein convected to the membrane. This is consistent with the lack of any measurable affect of stirring on either the pressure drop or protein transmission in the skin-side down orientation (data not shown), in contrast to the results obtained with the skin-side up (Figure 4).

The bottom panel of Figure 7 shows the corresponding data for the total resistance for experiments performed with the membrane oriented so that the skin side faces the feed (results from Figures 1 and 3). In contrast to the data with the skin-side down, the results with the skin-side up show a strong dependence on the mode of operation (constant flux versus constant pressure) as well as on the magnitude of the pressure or flux used in the experiment. The runs with constant pressure filtration were characterized by a rapid initial increase in resistance, with the rate of increase being greatest at the highest pressure. The runs with constant flux began with a relatively slow and steady increase in the total resistance, but the resistance then increased rapidly at longer filtration times. It is important to note that all of the classical fouling models (complete pore blockage, intermediate pore blockage, pore constriction, and cake filtration) predict that the total resistance should depend only on the total mass of protein convectively carried to the membrane, which is directly proportional to the cumulative filtrate volume for experiments performed at a constant feed concentration. The strong dependence on both the operating conditions and

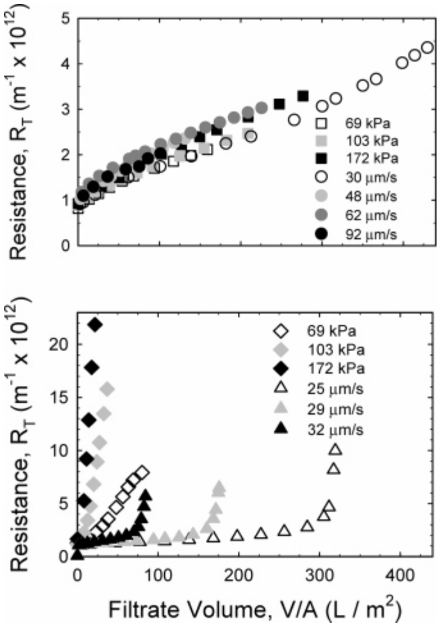

Figure 7. Total resistance as a function of the cumulative filtrate volume per unit membrane area for constant flux and constant pressure operation using the skin-side down (top panel) and skin-side up (bottom panel) orientations.

operating mode in the skin-side up orientation is consistent with the large contribution of osmotic pressure to the observed behavior, with greater osmotic pressures obtained at higher filtrate flux due to the greater degree of concentration polarization.

The initial rate of membrane fouling, $dR_{total}/d(V/A)$, was evaluated from the slope of the data in Figure 7 (for $t < 600$ s), with the results plotted in Figure 8 as a function of the initial filtrate flux. The data obtained with the skin-side down orientation are essentially constant at a value of $1.4 \pm 0.3 \times 10^{13}$ m^{-2} independent of the initial filtrate flux for experiments performed using both constant pressure and constant filtrate flux. In contrast, the results for the skin-side up orientation have a very strong dependence on the initial filtrate flux, with $dR_{total}/d(V/A)$ increasing by more than 2 orders of magnitude as the initial flux increases from 24 to 100 μm/s. The strong dependence of the initial rate of fouling on the filtrate flux may be due to the greater degree of concentration polarization, and thus the greater osmotic pressure difference, at high flux. In addition, there may be some compressibility of the fouling layer, with the specific resistance of the protein deposit increasing at high pressures.

Discussion

The design of virus filtration processes is determined primarily by the system capacity, the total filtrate volume per unit membrane area that can be collected before the flux decreases to an unacceptably low value (for constant pressure operation) or before the pressure increases to its maximum value (for constant flux operation). The data obtained in this study provide the first quantitative results for the effects of membrane orientation and operating conditions on the system capacity and fouling of the Viresolve 180 membrane using cysteinylated bovine serum albumin as a model protein product. The results clearly demonstrate that the system capacity with the skin-side down is significantly larger than that with the skin-side up. For example, the system capacity for constant pressure operation at 103 kPa with the skin-side down is more than 240 L/m^2 compared to only 13 L/m^2 with the skin-side down. Similar differences were seen with constant flux operation, with the system capacity increasing by more than a factor of 2 at a flux of approximately 30 μm/s. These differences appear to be due to the different fouling mechanisms in the two orientations; the flux decline (or pressure rise) in the skin-side up orientation is dominated by osmotic pressure effects, whereas the same phenomena are determined primarily by irreversible fouling within the porous substructure when the membrane is used in the skin-side down orientation.

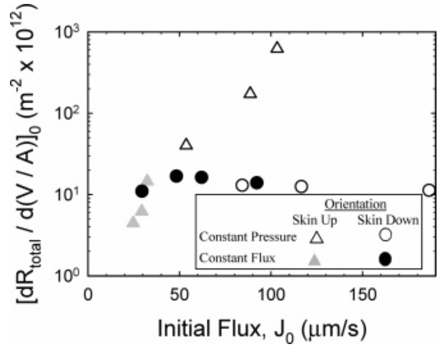

Figure 8. Initial rate of fouling as a function of the initial filtrate flux for operation with the skin-side down and skin-side up.

When the Viresolve 180 membrane was used with the skin side facing downstream (away from the feed), the total resistance was determined entirely by the filtrate volume per unit membrane area, which is proportional to the mass of protein convected to the membrane. The rate of fouling in this orientation was essentially independent of both the mode of operation (constant pressure or constant flux) and the magnitude of the flux/pressure. Scanning electron micrographs suggest that there is little protein fouling on the membrane substructure, with protein deposition occurring throughout the membrane thickness.

In contrast, the fouling behavior with the skin-side up orientation was a strong function of the operating mode and filtration conditions. Filtration at constant pressure gave a rapid but fairly steady rise in resistance, with the system capacity decreasing as the pressure increased due to the large degree of fouling associated with the high initial flux at high pressures. Experiments at constant flux were characterized by a slow initial rise in pressure, with the pressure subsequently increasing very rapidly above a critical filtration volume (or pressure). This rapid rise in pressure is apparently due in large part to concentration polarization effects; the transmembrane pressure and filtrate concentration both decreased rapidly upon stirring the solution above the membrane. The net result is that the system capacity for constant flux operation with the skin-side up is a very strong function of the value of the filtrate flux. Reducing the flux from 32 to 25 μm/s causes an almost 4-fold increase in the capacity. These results provide important insights into the factors governing the filtrate flux and system capacity during virus filtration, including the critical role of the membrane orientation (skin-side up or skin-side down).

Acknowledgment

The authors would like to thank Millipore Corporation for donating the membranes used in this study and Pratik Mankidy for his assistance in collecting the SEM images.

Note Added after ASAP Publication. There were minor errors in the version published ASAP on May 11, 2006; the corrected version was published May 23, 2006.

References and Notes

(1) Burnouf, T.; Radosevich, M. Nanofiltration of plasma-derived biopharmaceutical products. *Haemophilia* **2003**, *9*, 24−37.

(2) Carter, J.; Lutz, H. An overview of viral filtration in biopharmaceutical manufacturing. *Eur. J. Parenter. Sci.* **2002**, *7*, 72−78.

(3) Levy, R.; Phillips, M.; Lutz, H. Filtration and the removal of viruses from biopharmaceuticals. In *Filtration in the Biopharmaceutical Industry*; Meltzer, T., Jornitz, M., Eds.; Marcel Dekker: New York, 1998; pp 619−646.

(4) Brough, H.; Antoniou, C.; Carter, J.; Jakubik, J.; Xu, Y.; Lutz, H. Performance of a novel Viresolve NFR virus filter. *Biotechnol. Prog.* **2002**, *18*, 782−795.

(5) Marshall, A.; Munro, P.; Tragardh, G. The effect of protein fouling in microfiltration and ultrafiltration on permeate flux, protein retention and selectivity: A literature review. *Desalination* **1993**, *91*, 65−108.

(6) Belfort, G.; Davis, R.; Zydney, A. The behavior of suspensions and macromolecular solutions in cross-flow microfiltration. *J. Membr. Sci.* **1994**, *96*, 1−58.

(7) Badmington, F.; Wilkins, R.; Payne, M.; Honig, S. Vmax testing for practical microfiltration train scale-up in biopharmaceutical processing. *Pharm. Technol.* **1995**, *9*, 64−86.

(8) Higuchi, A.; Komuro, A.; Hirano, K.; Yoon, B.-O.; Hara, M.; Hirasaki, T.; Nishimoto, M.; Yokogi, M.; Manabe, S. Permeation of γ-globulin through microporous membranes in the presence of trace DNA. *J. Membr. Sci.* **2001**, *186*, 9−18.

(9) Higuchi, A.; Nemoto, M.; Koyama, H.; Hirano, K.; Yoon, B.-O.; Hara, M.; Yokogi, M.; Manabe, S. Enhanced microfiltration of γ-globulin solution upon treatment of NaCl addition and/or DNase digestion. *J. Membr. Sci.* **2002**, *210*, 369−378.

(10) Higuchi, A.; Kyokon, M.; Murayama, S.; Yokogi, M.; Hirasaki, T.; Manabe, S. Effect of aggregated protein sizes on the flux of protein solution through microporous membranes. *J. Membr. Sci.* **2004**, *236*, 137−144.

(11) DiLeo, A.; Allegrezza, A.; Builder, S. High-resolution removal of virus from protein solutions using a membrane of unique structure. *Bio/Technology* **1992**, *10*, 182.

(12) Robertson, B.; Zydney, A. Protein adsorption in asymmetric ultrafiltration membranes with highly constricted pores. *J. Colloid Interface Sci.* **1990**, *134*, 563−575.

(13) Maruyama, T.; Katoh, S.; Nakajima, M.; Nabetani, H. Mechanism of bovine serum albumin aggregation during ultrafiltration. *Biotechnol. Bioeng.* **2001**, *75*, 233−238.

(14) Kelly, S.; Zydney, A. Mechanisms for BSA fouling during microfiltration. *J. Membr. Sci.* **1995**, *107*, 115−127.

(15) Ho, C.-C.; Zydney, A. Effect of membrane morphology on the initial rate of protein fouling during microfiltration. *J. Membr. Sci.* **1999**, *155*, 261−275.

(16) Ho, C.-C.; Zydney, A. Effect of membrane morphology on system capacity during normal flow microfiltration. *Biotechnol. Bioeng.* **2003**, *83*, 537−543.

(17) Kelly, S.; Zydney, A. Effects of intermolecular thiol-disulfide interchange reactions on BSA fouling during microfiltration. *Biotechnol. Bioeng.* **1994**, *44*, 972−982.

(18) Smith, K.; Colton, C.; Merrill, E.; Evans, L. Convective transport in a batch dialyzer: determination of the true membrane permeability from a single measurement. *AIChE Symp. Ser.* **1968**, *64*, 45.

(19) Opong, W.; Zydney, A. Diffusive and convective protein transport through asymmetric membranes. *AIChE J.* **1991**, *37*, 1497−1510.

(20) Vilker, V.; Colton, C.; Smith, K. The osmotic pressure of concentrated protein solutions: effect of concentration and pH in saline solutions of bovine serum albumin. *J. Colloid Interface Sci.* **1981,** *79*, 548−566.

(21) Bohonak, D.; Zydney, A. Compaction and permeability effects with virus filtration membranes. *J. Membr. Sci.* **2005,** *254*, 71−79.

(22) Jonsson, G. E.; Wenten, I. G. Control of concentration polarization, fouling and protein transmission of microfiltration processes within the agro-based industry. *Proceedings ASEAN-EC Workshop on Membrane Technology in Agro-based Industry* **1994**, 157−166.

Accepted for publication April 11, 2006.

BP050350V